

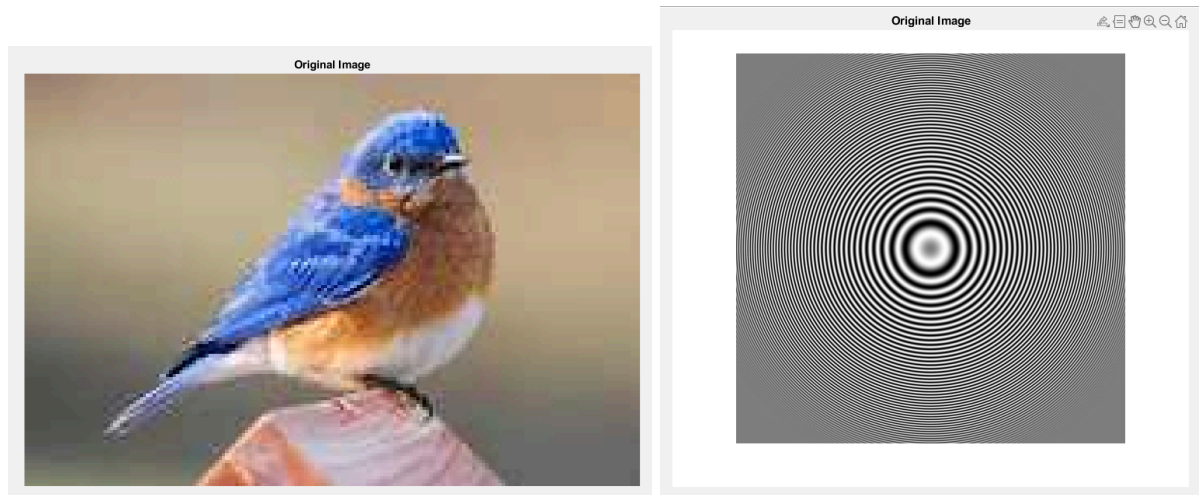
COMPE 510 – PA 5 Report

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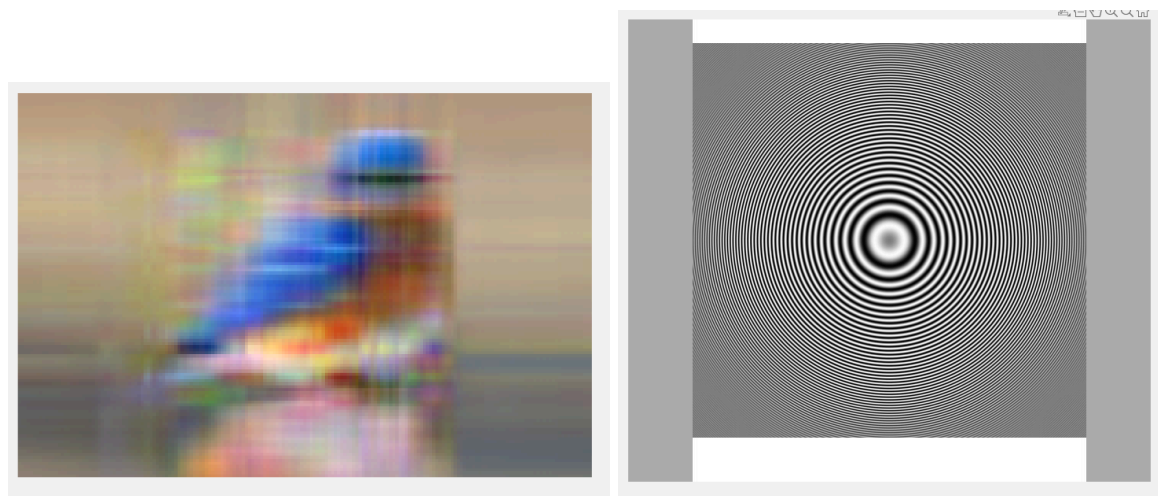
Results

Part 2: Perform PCA to Compress the Image

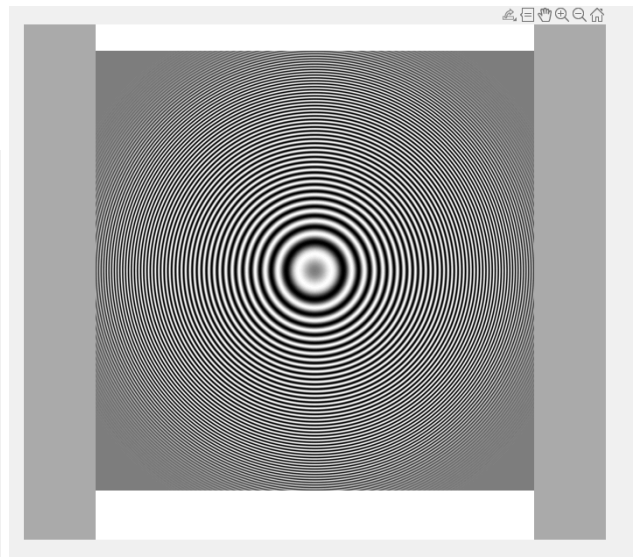
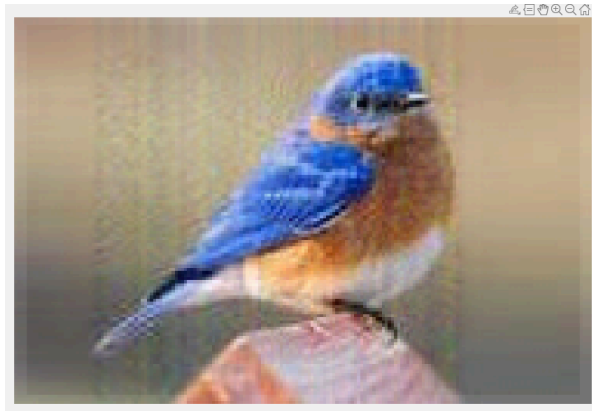
Original Images



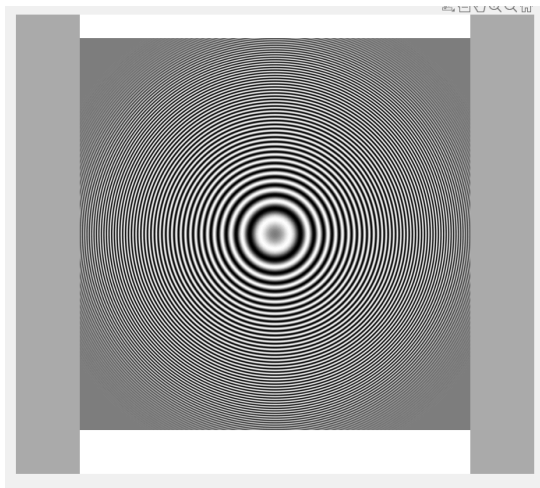
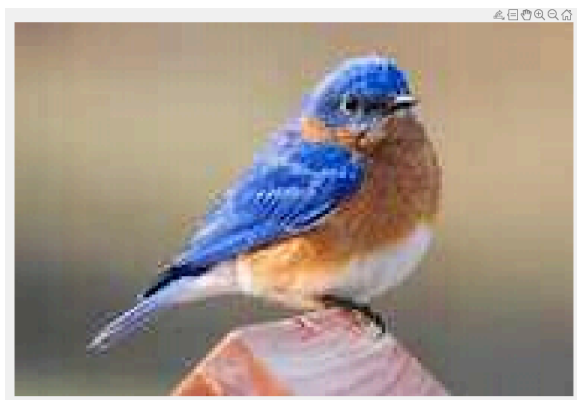
Compressed Images, $K = 5$



Compressed Images, $K = 30$



Compressed Images, $K = 100$



Questions:

1. The value of K corresponds to the image quality and compression, with higher values of K resulting in higher image quality compared to lower values. Images are noticeably blurrier for lower K values, as seen in the bluebird images.
2. PCA compared to JPEG is worse at image compression, but better at capturing important patterns in data. For machine learning purposes, PCA is better than JPEG since it reduces the dimensionality of images which is user-definable (picking the number of principal components to keep).
3. I picked an image with next to no color variation, but what looks like more texture. After applying PCA, the compression results were nearly identical to the original image. Compared to the original bird image, there was no noticeable blurring effect.

Implementation

Implementing PCA went well – there was an issue where the images turned out significantly darker than the original, but after adding the mean the colors were more inline with the original.